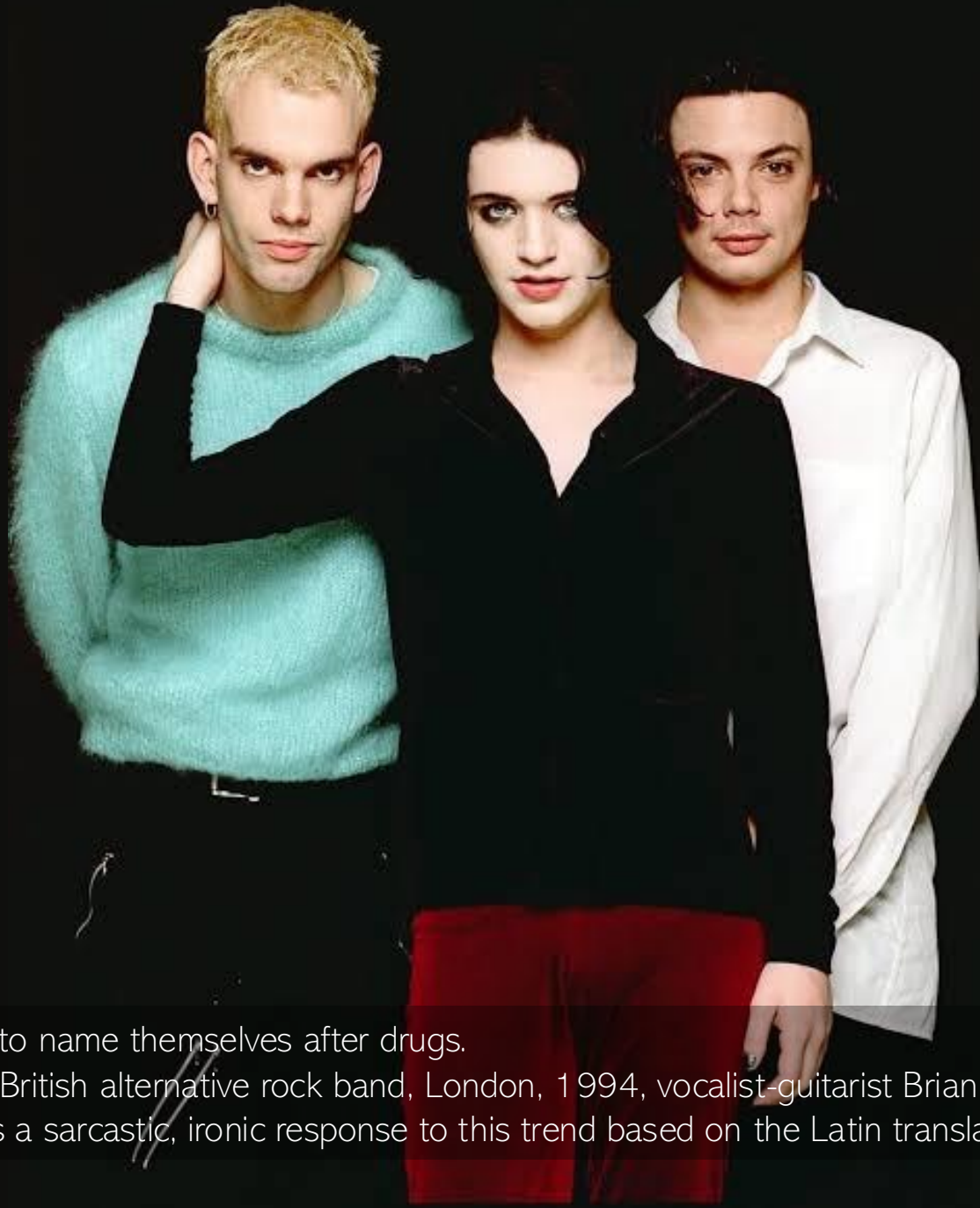




In the 1990s, it was a cliché for bands to name themselves after drugs. Placebo (/plə'si:bʊ/ plə-SEE-boh), a British alternative rock band, London, 1994, vocalist-guitarist Brian Molko and bassist-guitarist Stefan Olsdal. Placebo, referring to a 'dummy' drug, as a sarcastic, ironic response to this trend based on the Latin translation meaning "*I shall please*"

🎉 It's time to celebrate IEEE Day 2026! From 4–17 October, the IEEE Education Society is joining global celebrations of innovation, technology, and the educators shaping the future of engineering.

📄 Exclusive IEEE Day Offer: Joining or renewing? Student Members and First-Time Professionals can unlock special membership discounts and promotional offers during this two-week celebration!

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Explore opportunities
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COLLABORATE
Build a stronger
IEEE community



MAKE AN IMPACT
Be part of a better
tomorrow

EVENT SCHEDULE

 Date Tuesday, October 6	 Time 12:00 p.m. – 2:00 p.m.
 Venue ELW Lobby University of Victoria	

What to Expect

-  Slide show introducing IEEE, IEEE Victoria and our Chapters/AG
-  Student branch information desk with IEEE birthday cake
-  Pizza lunch for members and guests
-  Open house of the IEEE McNaughton Center
-  Field tour of the world's largest LEO satellite network testbed





Overview

Bookmarks

Course Schedule

1

Table of Contents

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Discussion

1

Official Course Outline

1

Course Outline (Syllabus)

3. Solution Design and Implementation



Print

Settings

Add dates and restrictions...

Add a description...

Upload / Create



Existing Activities



Bulk Edit

3. Solution Design & Implementation



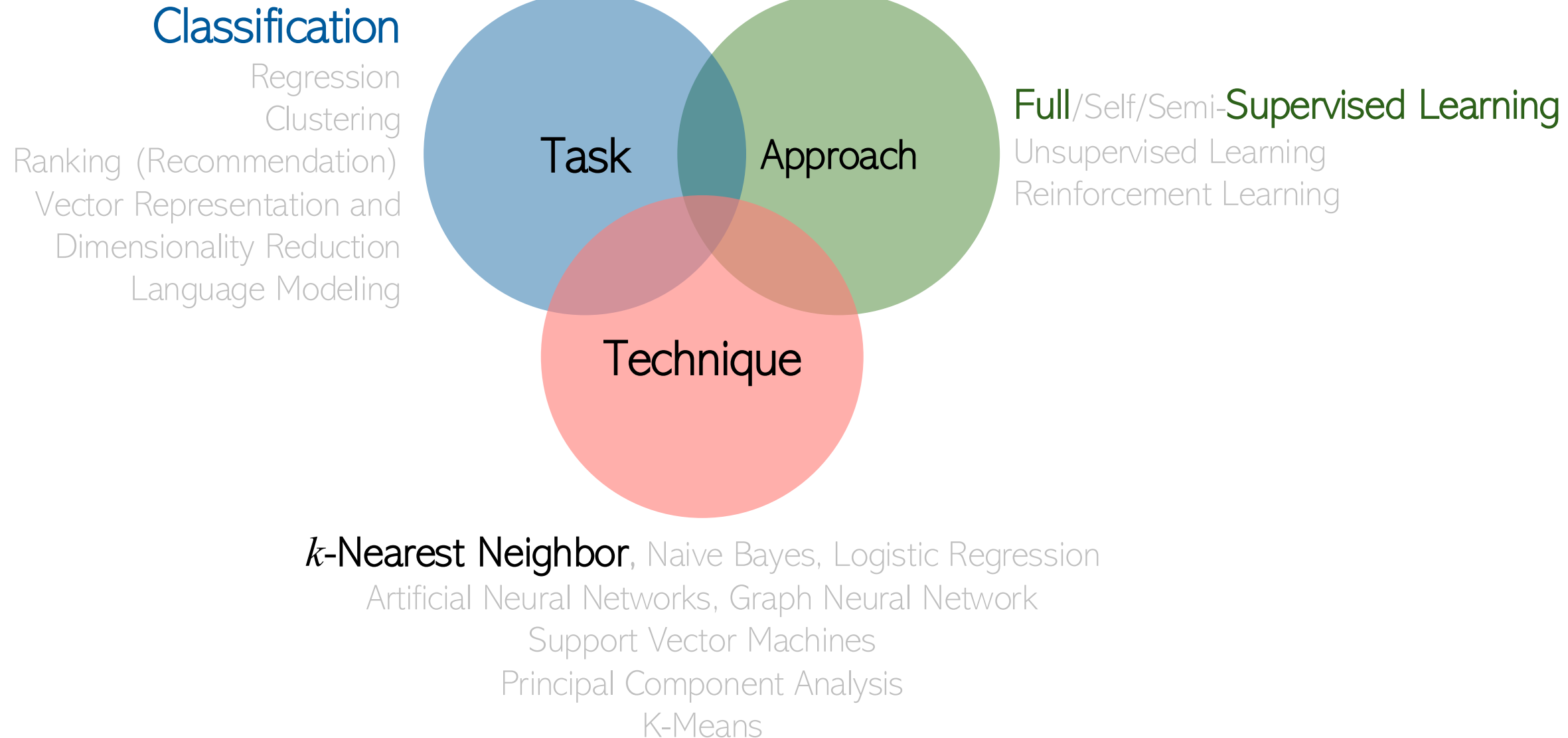
Assignment



Due November 5 at 6:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

4 Outline: W02: Topics Start



Outline: W03-W04

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

Let's zoom in ...

Hyperparameters ...

Distance ...

Embedding ...

IIII) Evaluation

6

k -Nearest Neighbor: Domain Challenges

How to map a real-world observation to a point in Euclidian space?

People

Animals (Birds, ...)

Objects (Houses, Cars, ...)

Image (Biometric Iris)

Audio (Voice, Music)

Video (Movies)

Text (Words, Sentence, Docs)

Timeseries

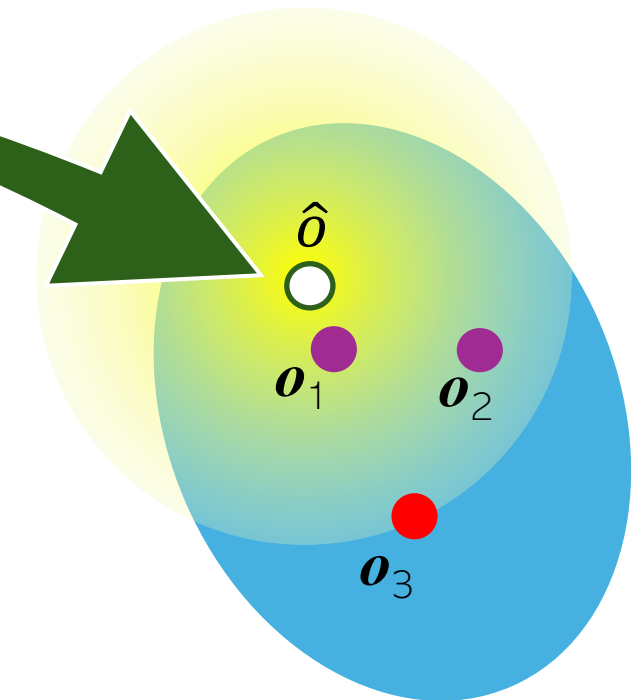
Genes

Spatial (GIS)

Light Bulbs

...

Encoding
Embedding
 $e(o)$

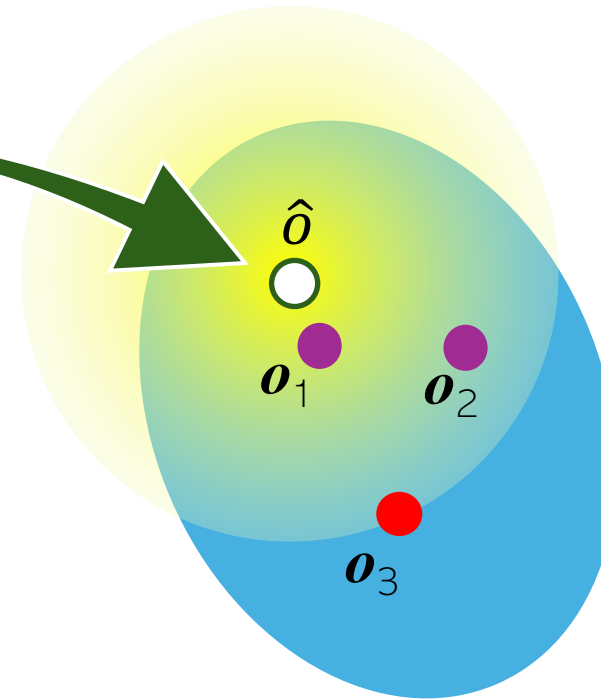


Euclidian Space
(Embedding Space)

7 k -Nearest Neighbor: Text

Words
Sentences
Docs
Social Posts
Books
Webpages
...

$e(o)$



Euclidian Space
(Embedding Space)

8

k -Nearest Neighbor: Text: Word Space $\rightarrow$ Euclidian Space

Real-World Problem: Named Entity Annotator

Task: Classification $\mathbf{C} = \mathbf{Entities} = \{\text{Person, Organization, Event, Food, Animal}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(w, c)\}$, $\mathbf{U}^* = \{(\hat{w}, f(\hat{w}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{w}, w) \approx d(e(\hat{w}), e(w)) \rightarrow e: \mathbf{U} = \mathbf{Vocabs} \rightarrow \mathbb{R}^{n=1}$

9

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=1}$

Indexing $\rightarrow$ Enumerate words in a language's dictionary (vocabulary set)



$$e(\hat{w}: \text{'aardvark'})=0$$



$$e(w: \text{'apple'})=1050$$



$$e(w: \text{'zyzzyva'})=|V|$$

Does $e(.)$ preserve “real-world” proximity to “spatial” proximity? **No!**

$$\begin{aligned} |e(\text{'aardvark'}) - e(\text{'apple'})| &< |e(\text{'aardvark'}) - e(\text{'zyzzyva'})| \\ |0 - 1050| &< |0 - 520,000| \end{aligned}$$

‘aardvark’ is closer to ‘apple’ vs. ‘zyzzyva’!

‘aardvark’ is tech company!

10

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: **Distributional Hypothesis**

Location of the words does *not* matter.

More cooccur in the same context

- More similar semantic
- Closer meaning
- Closer the distance!

Linguistic Semantic (Real-World) Proximity

- Cooccurrence Context in H1
- **Distributional Semantics**
- Geometric Proximity

Computational Leverage

not good
dislike
bad
worst
incredibly bad
worse

amazing
terrific
very good
incredibly good
fantastic
wonderful
nice
good

12

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

(1) Pick a text corpus (may or may not the same as $\mathbf{O}^*$)

(2) Define Context =

Fixed: Window of $\pm m$ words

Dynamic: Phrase, Sentence, Social Post, Paragraph, Page, Chapter, Book

(3) Count Word-Word Cooccurrences

13

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Corpus: Wikipedia articles
 Context: Window of ± 4 words

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	
strawberry	0	...	0	0	1	60	19	
digital	0	...	1670	1683	85	5	4	
information	0	...	3325	3982	378	5	13	

 $1 \times |V|$ $1 \times |V|$ $1 \times |V|$ $1 \times |V|$

Word-Word Matrix
 Term-Term Matrix
 Word-Context Matrix

$$|V| \times |V|$$

14

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Small (± 1) or Large (± 1000) Context? Should be tuned ...

The smaller the context,

The less chance two words cooccur

More sparsity, less noise in semantic similarity

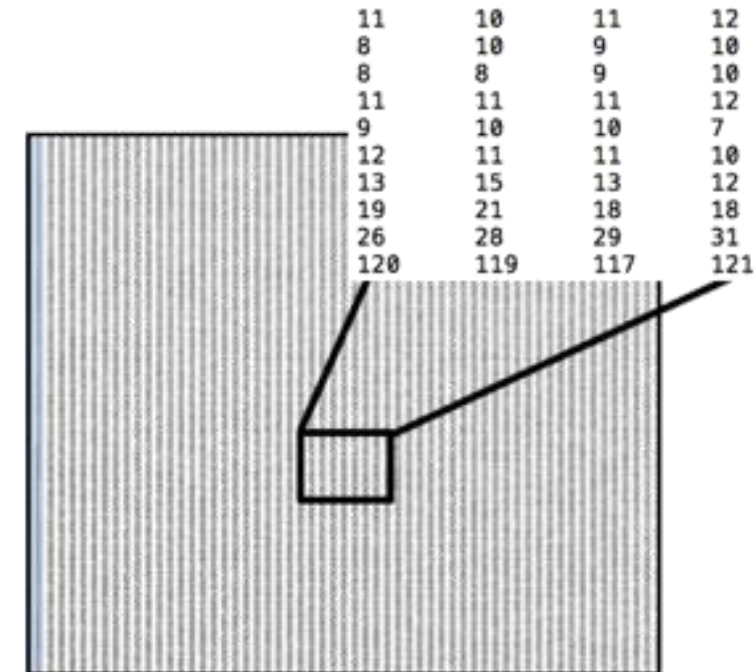
Conservative on semantic proximity

The larger the context,

The more chance two words cooccur

Less sparsity, more noise in semantic similarity

Liberal on semantic proximity



15

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Word-Word (Term-Term) (Word-Context) Matrix

(1) Pick a text corpus (may or may not be the same as $\mathbf{O}^*$)

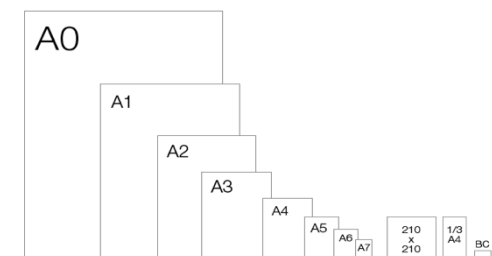
(2) Define Context =

Fixed: Window of $\pm m$ words

Dynamic: Phrase, Sentence, Social Post, Paragraph, Page, Chapter, Book

Document $\rightarrow$ a single line/file

(3) Count Word-Word Cooccurrences: all words in a line/file cooccur



16

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|\mathbf{V}|}$

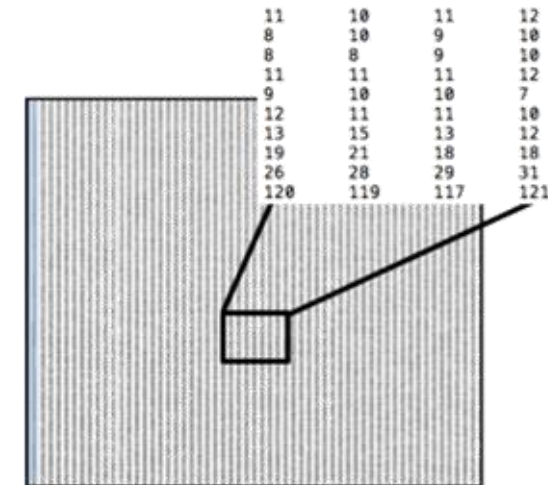
What happens to k -NN complexity? We already saw that $n \times |\mathbf{O}^*|$

$n = |\mathbf{V}| \simeq 520,000$ (size of online oxford dictionary)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$

For 1 unseen observation: $|520,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to a manageable size?



17

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	the	
cherry	0	...	2	8	9	442	25	205	$1 \times V $
strawberry	0	...	0	0	1	60	19	324	$1 \times V $
the	452	...	504	560	800	759	398	0	$1 \times V $

$|V| \times |V|$

Stop words, a static finite set of **non-informative** words, e.g., the, a, an, as
 $\rightarrow$ Filtering them shrinks the matrix horizontally/vertically



A friend to all
is a friend to none.

- Aristotle

Corollary by Hossein: "having such friend is not a value."

18

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	find	
cherry	0	...	2	8	9	442	25	120	$1 \times V $
strawberry	0	...	0	0	1	60	19	524	$1 \times V $
find	452	...	504	560	800	759	398	0	$1 \times V $

 $|V| \times |V|$

But we need a dynamic list depending on the domain, e.g., lol, :), :D,

→ Dynamic set of **non-discriminative** words

→ Filtering them shrinks the matrix horizontally/vertically



A friend to all
is a friend to none.

- Aristotle

Corollary by Hossein: "having such friend is not a value."

19

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	... find
cherry	0	...	2	8	9	442	25	120 $\rightarrow$ 0
strawberry	0	...	0	0	1	60	19	524 $\rightarrow$ 0
find	452 $\rightarrow$ 0	...	504 $\rightarrow$ 0	560 $\rightarrow$ 0	800 $\rightarrow$ 0	759 $\rightarrow$ 0	398 $\rightarrow$ 0	0

$$|V| \times |V|$$

The more friend you have, the less valuable you are

The more cooccur with other words, the less accountable

(**find**, **aardvark**)=452 but also cooccur with many other words.

How many $\rightarrow$ Context Frequency (CF) $\rightarrow$ Row/Column Nonzero Counts $\rightarrow$ 6

Discount the value by CF $\rightarrow \frac{452}{CF=6} = 452 \times \frac{1}{6}$

inverse Context Frequency (i CF)

20

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	find
cherry	$0 \times \frac{1}{6}$	...	$2 \times \frac{1}{6}$	$8 \times \frac{1}{6}$	$9 \times \frac{1}{6}$	$442 \times \frac{1}{6}$	$25 \times \frac{1}{6}$	$120 \times \frac{1}{6}$
strawberry	$0 \times \frac{1}{4}$	...	$0 \times \frac{1}{4}$	$0 \times \frac{1}{4}$	$1 \times \frac{1}{4}$	$60 \times \frac{1}{4}$	$19 \times \frac{1}{4}$	$524 \times \frac{1}{4}$
find	$452 \times \frac{1}{7}$	...	$504 \times \frac{1}{7}$	$560 \times \frac{1}{7}$	$800 \times \frac{1}{7}$	$759 \times \frac{1}{7}$	$398 \times \frac{1}{7}$	$0 \times \frac{1}{7}$

$$CF(\mathbf{cherry})=6$$

$$CF(\mathbf{strawberry})=4$$

$$CF(\mathbf{find}, \mathbf{aardvark})=7$$

Discounts by
inverse Context Frequency



21

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|\mathbf{V}|}$

What happens to k -NN complexity? We already saw that $n \times |\mathbf{O}^*|$

$n = |\mathbf{V}| \simeq 520,000$ (size of online oxford dictionary)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$

For 1 unseen observation: $|520,000| \times |\mathbf{O}^*|$

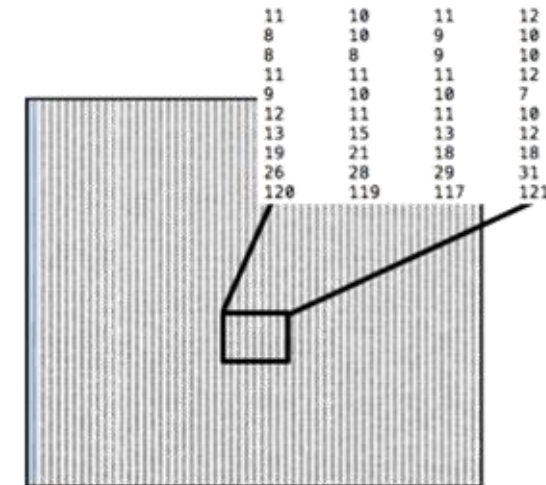
Can we reduce the $|\mathbf{V}|$ to a manageable size?

$n = |\mathbf{V}| \simeq 120,000$ (depending on the domain)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$

For 1 unseen observation: $|120,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to an **even more** manageable size?



22 k -Nearest Neighbor: Text

Word Space

- Birds of a Feather Flock Together (*“but where?”*)
- Birds of a Feather Flock **in the same/similar places!**

Instead of counting cooccurrences,

- Counting the occurrences in **places** of cooccurrences ...
- Number of **places** are way less than words ...

The students of this class will cooccur with students of previous and next classes. *In a general building, not informative.*

The students of this class will cooccur with students of previous and next classes. *In a faculty building, informative.*

Hossein (ECS: 6, EOW: 0, BEC: 0) days a week

Ali (ECS: 5, EOW: 2, BEC: 0) days a week

Sharon (ECS: 0, EOW: 0, BEC: 5) days a week

23 k -Nearest Neighbor: Text

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|} \ll |\mathbf{V}|$

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each contains the number of times the (row) word occurs in the (column) document.

$|\mathbf{V}| \times |\mathbf{D}|$

Context = Document

Word-Word Matrix $\rightarrow$ Word-Doc Matrix = $|\text{Vocabs}| \times |\text{Documents}|$

24 k -Nearest Neighbor: Text

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|} \ll |\mathbf{V}|$

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each contains the number of times the (row) word occurs in the (column) document.

$|\mathbf{V}| \times |\mathbf{D}|$

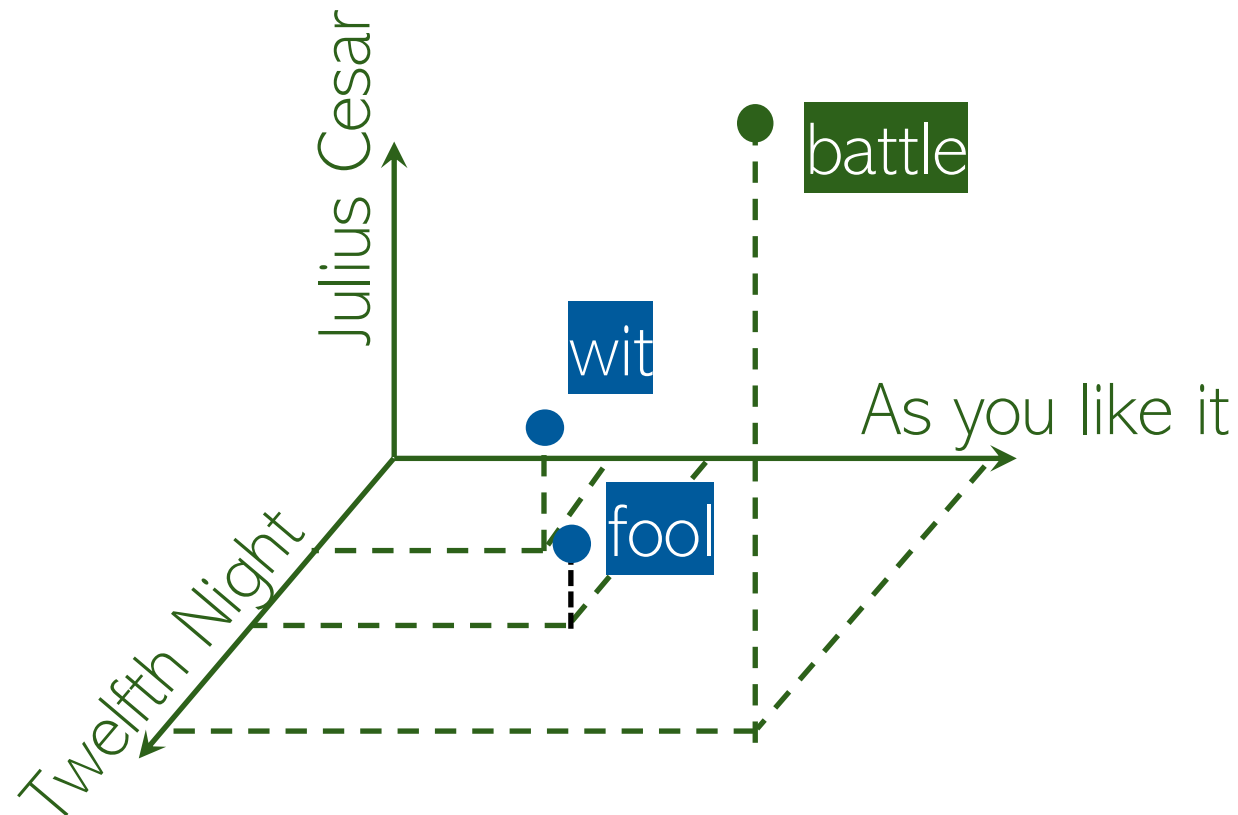
battle vs. fool, wit

battle and fool are *not* used in the same play (context), so different semantics
 fool and wit are used in the same play (context), so similar semantics

25 k -Nearest Neighbor: Text

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|} \ll |\mathbf{V}|$

Word are points, distributed in Document Space



26 k -Nearest Neighbor: Text

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|} \ll |\mathbf{V}|$

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

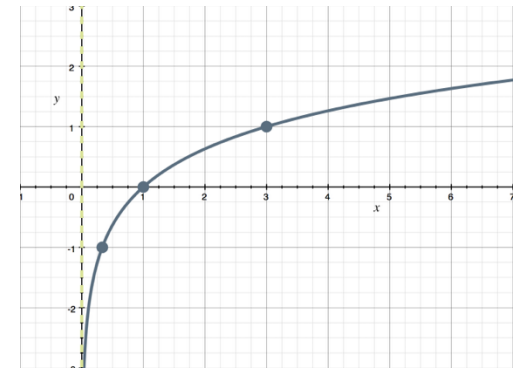
$|\mathbf{V}| \times |\mathbf{D}|$

Is 'good' discriminative? Informative? **No!**
 We already discuss how to filter these words out ...

27 k -Nearest Neighbor: Text

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|} \ll |\mathbf{V}|$

$$\text{idf}_t = \left(\frac{1}{\text{df}_t} \right) \xrightarrow{\text{Smoother Discount}} \log_{10} \left(\frac{N}{\text{df}_t} \right)$$



where N is the total number of documents in the collection

Word	df	idf
Romeo	1	1.57
fool	36	0.012
good	37	0

k -Nearest Neighbor: Text: TF- i DF

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|} \ll |\mathbf{V}|$

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

$|\mathbf{V}| \times |\mathbf{D}|$

Figure 6.8 A tf-idf weighted term-document matrix for four words in four Shakespeare plays, using the counts in Fig. 6.2. For example the 0.049 value for *wit* in *As You Like It* is the product of $\text{tf} = \log_{10}(20 + 1) = 1.322$ and $\text{idf} = .037$. Note that the idf weighting has eliminated the importance of the ubiquitous word *good* and vastly reduced the impact of the almost-ubiquitous word *fool*.

29

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|}$

What happens to k -NN complexity? We already saw that $n \times |\mathbf{O}^*|$

$n = |\mathbf{V}| \simeq 520,000$ (size of online oxford dictionary)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$, for 1 unseen observation: $|520,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to a manageable size?

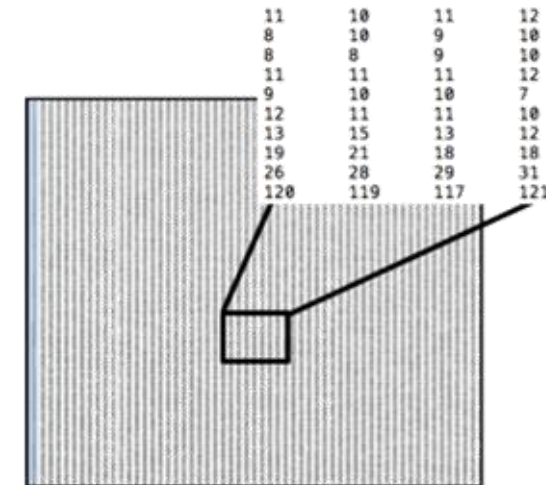
$n = |\mathbf{V}| \simeq 120,000$ (depending on the domain)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$, for 1 unseen observation: $|120,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to an **even more** manageable size?

$n = |\mathbf{D}| \simeq 2,000$ (depending on the domain)

k -NN $\rightarrow |\mathbf{D}| \times |\mathbf{O}^*|$, for 1 unseen observation: $|2,000| \times |\mathbf{O}^*|$



k -Nearest Neighbor: Document Space $\rightarrow$ Euclidian Space

$\mathbb{R}^{n=|V|} \leftarrow \text{Word Space} \leftarrow \text{Document Space}$

	As You Like It	Twelfth Night	Julius Caesar	Henry V	
battle	1	0	7	13	$ V \times D $
good	114	80	62	89	
fool	36	58	1	4	
wit	20	15	2	3	
	$ V \times 1$	$ V \times 1$	$ V \times 1$	$ V \times 1$	

Transpose (T): Rotate 90°

Documents are points, distributed in the word space!

If we encode/embed buildings, they would be distributed over students' occurrence in buildings ...

ECS: (Hossein: 6, Ali: 5, Sharon: 0)

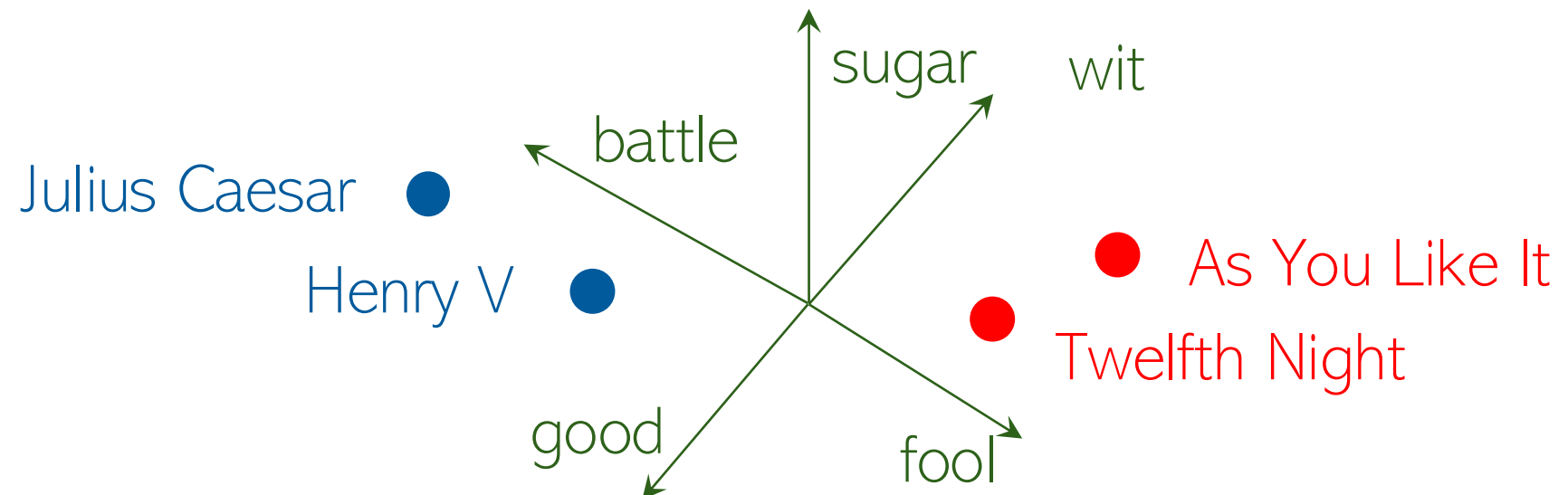
BEC: (Hossein: 0, Ali: 0, Sharon: 5)

31

 k -Nearest Neighbor: Document Space $\rightarrow$ Euclidian Space $\mathbb{R}^{n=|\mathbf{V}|} \leftarrow \text{Word Space} \leftarrow \text{Document Space}$

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

$|\mathbf{V}| \times |\mathbf{D}|$



32

 k -Nearest Neighbor: Document Space $\rightarrow$ Euclidian Space

[15]

✓ Os

```

1  import pandas as pd
2  from sklearn.feature_extraction.text import CountVectorizer
3
4  docs = [
5      "ongchoi is a popular leafy green vegetable",
6      "ongchoi is commonly cooked with garlic",
7      "people eat ongchoi in soups and stir fries"
8  ]
9
10 count_model = CountVectorizer()
11 X = count_model.fit_transform(docs)
12 words = count_model.get_feature_names_out()
13
14 # Print Document-Word matrix
15 X_df = pd.DataFrame(X.toarray(), index=[f"Doc {i+1}" for i in range(len(docs))], columns=words).transpose()
16 print(X_df.to_string())

```

✓

```

...      Doc 1  Doc 2  Doc 3
and          0      0      1
commonly     0      1      0
cooked       0      1      0
eat          0      0      1
fries        0      0      1
garlic       0      1      0
green        1      0      0
in           0      0      1
is           1      1      0
leafy        1      0      0
ongchoi      1      1      1
people       0      0      1

```

33

 k -Nearest Neighbor: Word-Document Matrix

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|}$

	As You Like It	Twelfth Night	Julius Caesar	Henry V	
battle	1	0	7	13	Word Point
good	114	80	62	89	
fool	36	58	1	4	
wit	20	15	2	3	
					$ \mathbf{V} \times \mathbf{D} $

Doc Point

$\mathbb{R}^{n=|\mathbf{V}|} \leftarrow$ Word Space $\leftarrow$ Document Space

k -Nearest Neighbor: Document Space $\rightarrow$ Euclidian Space

Real-World Problem: Document Topic Classification

Task: Classification $\mathbf{C}=\mathbf{Topics}=\{\text{Politic, Law, Sport, Art, Misc}\}$

Approach: Fully Supervised $\mathbf{O}^*=\{(d, c)\}, \mathbf{U}^*=\{(\hat{d}, f(\hat{d}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{d}, d) \approx d(e(\hat{d}), e(d)) \rightarrow e: \mathbf{U}=\mathbf{Docs} \rightarrow \mathbb{R}^n$

Document of same topics are using same/similar words!

Politic: *movement, campaign, party, rule, action, ...*

Law: *court, legal, lawsuit, contract, ...*

Sport: *field, event, play, ...*

Art: *art, paint, performance, ...*

35

k -Nearest Neighbor: Document Space $\rightarrow$ Euclidian Space

Real-World Problem: Review Sentiment Classification

Task: Classification $\mathbf{C} = \{\text{Happy, Not Happy}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(r, c)\}$, $\mathbf{U}^* = \{(\hat{d}, f(\hat{r}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{r}, r) \approx d(e(\hat{r}), e(r)) \rightarrow e: \mathbf{U} = \mathbf{Reviews} \rightarrow \mathbb{R}^n$

Reviews of same sentiment are using same/similar words!

Happy: *amazing, fantastic, ...*

Not Happy: *poor, bad, dirty, ...*

36

k -Nearest Neighbor: Document Space $\rightarrow$ Euclidian Space

Real-World Problem: Spam Email Detection

Task: Classification $\mathbf{C} = \{\text{Spam}, \text{Not Spam}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(m, c)\}$, $\mathbf{U}^* = \{(\hat{m}, f(\hat{m}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{m}, m) \approx d(e(\hat{m}), e(m)) \rightarrow e: \mathbf{U} = \mathbf{Emails} \rightarrow \mathbb{R}^n$

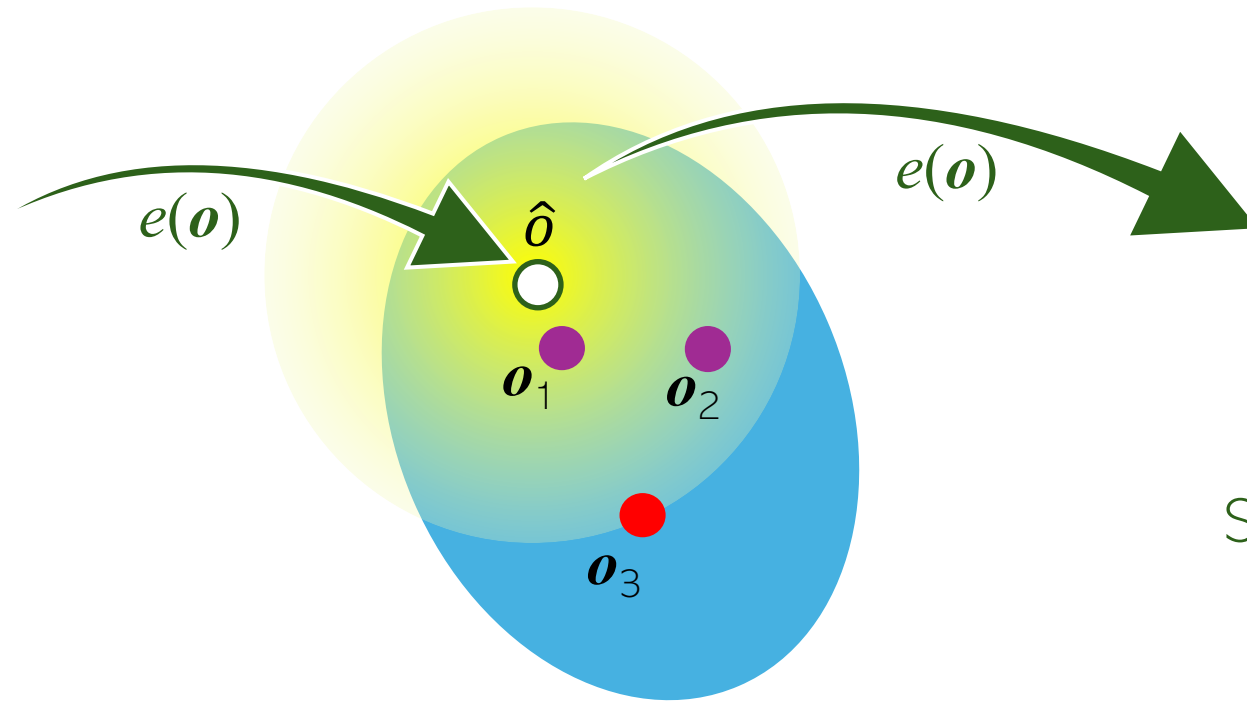
Spam emails are using same/similar words!

Spam: *free, winner, prize, offer, cash, bonus, ...*

Not Spam: *meeting, project, report, research, assignment, ...*

39

k -Nearest Neighbor: Transfer Learning: Single Point per Word

 k NN

Entity Annotator
Spam Detection
Sentiment Analysis
NLP Tasks

Apple (fruit) vs. Apple (company)

Python (snake) vs. Python (programming language)

Zyzzzyva (fly) vs. Zyzzzyva (magazine)

Placebo (dummy drug) vs. Placebo (music band)

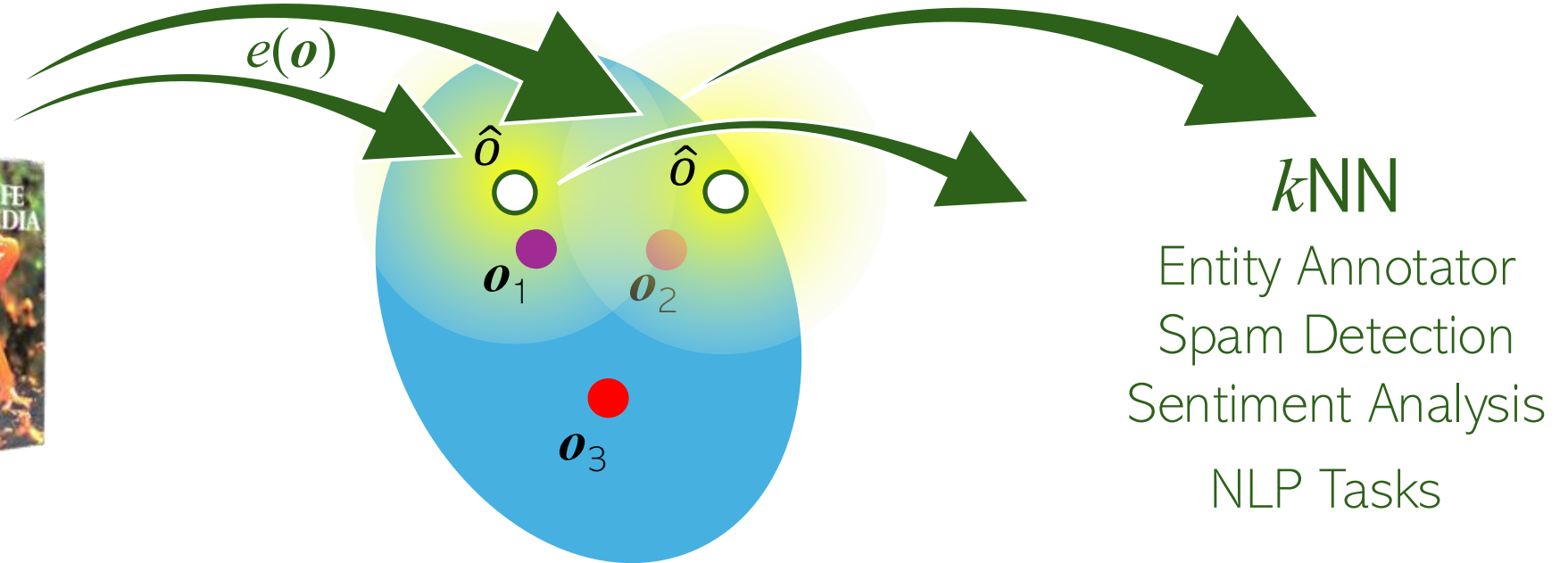


In the 1990s, it was a cliché for bands to name themselves after drugs. Placebo (/plə'si:bou/ plə-SEE-boh), a British alternative rock band, London, 1994, vocalist-guitarist Brian Molko and bassist-guitarist Stefan Olsdal. Placebo, referring to a 'dummy' drug, as a sarcastic, ironic response to this trend based on the Latin translation meaning "*I shall please*"

41

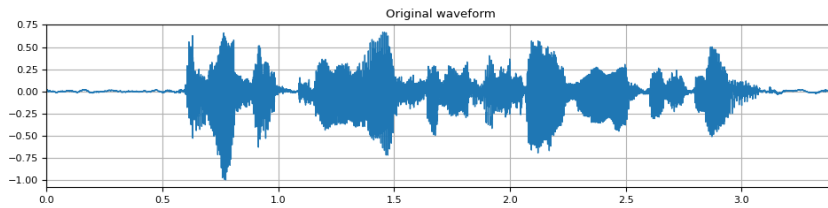
k -Nearest Neighbor: Transfer Learning

Can we have *multiple* points for a word?

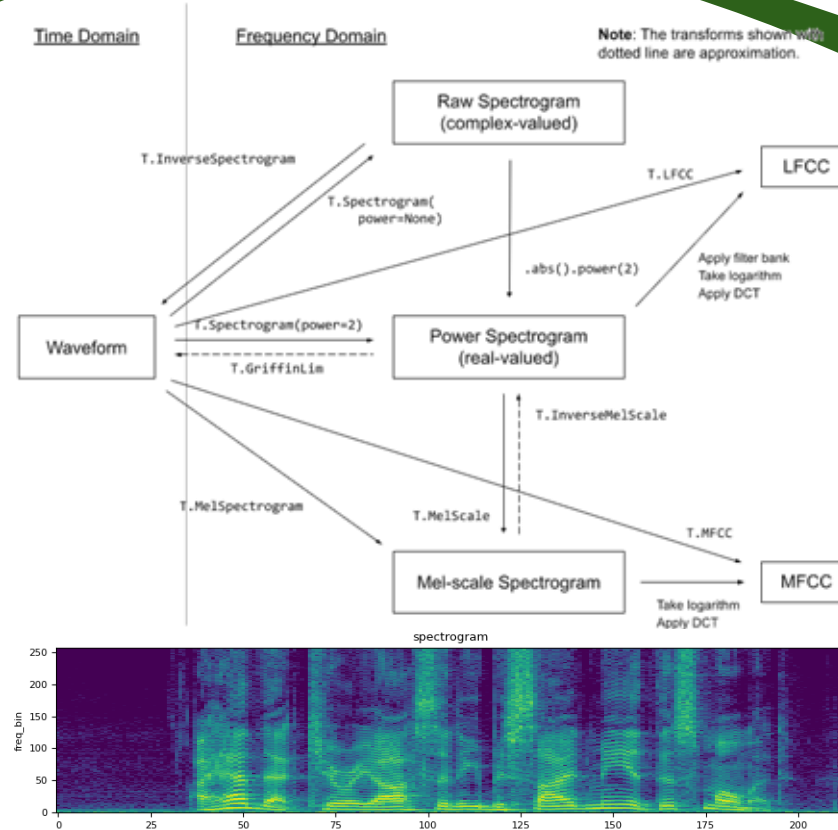


Apple (fruit) **and** Apple (company)
Python (snake) **and** Python (programming language)
Zyzzzyva (fly) **and** Zyzzzyva (magazine)
Placebo (dummy drug) **and** Placebo (music band)

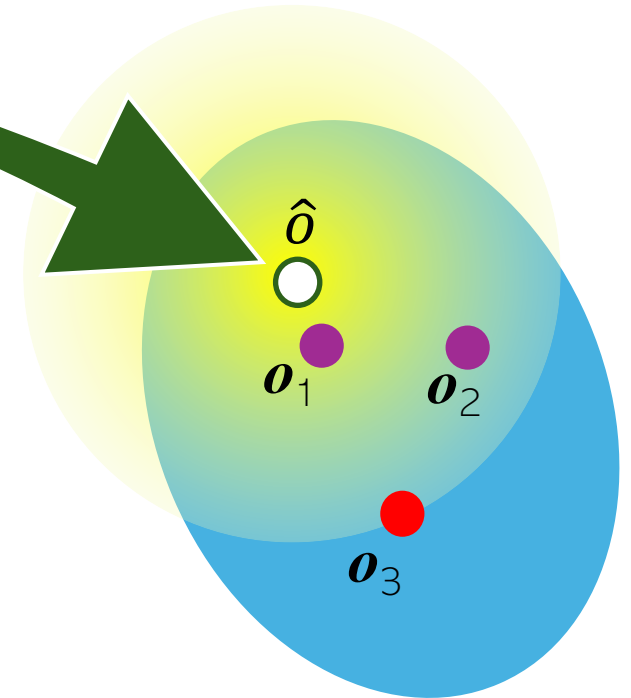
42

 k -Nearest Neighbor: Audio (Voice, Music, ...)

Wave in Time Space
 $\mathbb{R}^{n=|\text{Time}|}$

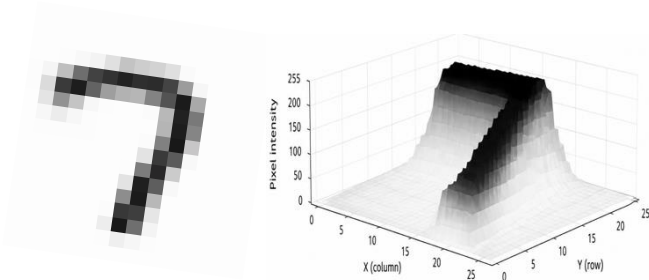


Fast Fourier Transform (FFT) in Frequency Space
 $\mathbb{R}^{n=|\text{FFT Coefficient}|}$

 $e(o)$ 

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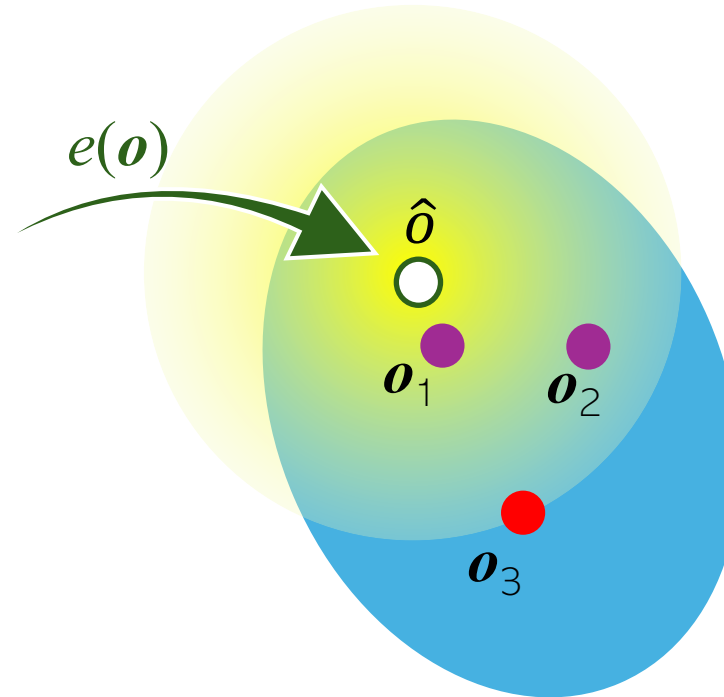
k -Nearest Neighbor: Image (MNIST Digits)



[	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	80	220	255	255	255	255	255	255	220	80	0	0	0]
[	0	0	0	0	0	0	0	0	90	255	255	0	0	0]
[	0	0	0	0	0	0	0	0	0	210	210	0	0	0]
[	0	0	0	0	0	0	0	0	140	255	90	0	0	0]
[	0	0	0	0	0	0	0	90	255	180	0	0	0	0]
[	0	0	0	0	0	0	80	255	200	0	0	0	0	0]
[	0	0	0	0	0	50	240	240	0	0	0	0	0	0]
[	0	0	0	0	0	180	255	90	0	0	0	0	0	0]
[	0	0	0	0	0	255	140	0	0	0	0	0	0	0]
[	0	0	0	0	140	255	0	0	0	0	0	0	0	0]
[	0	0	0	0	220	180	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0]

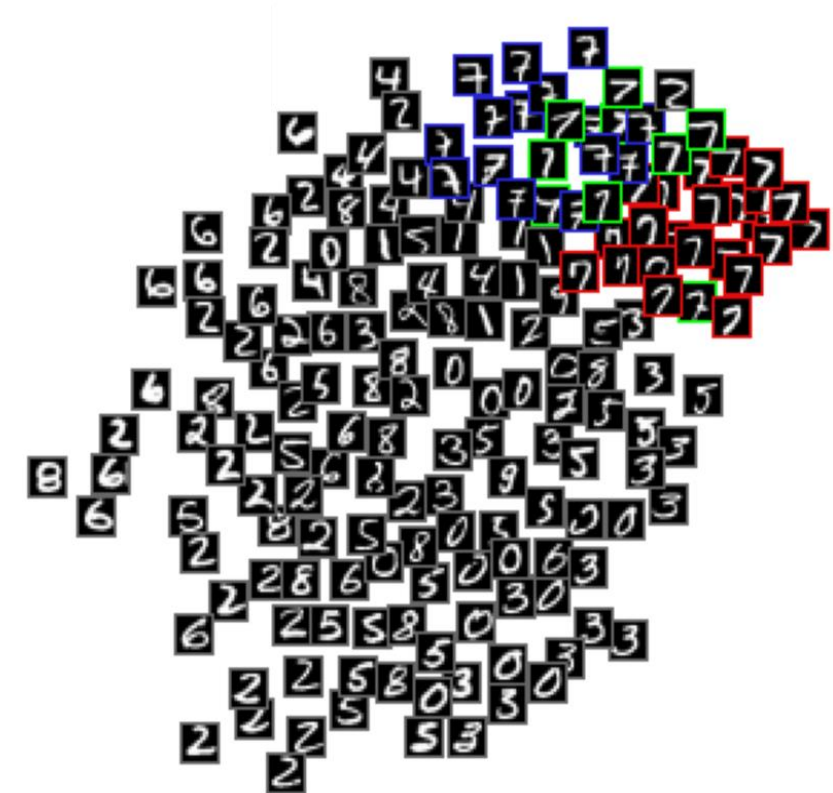
2D Wave in Grayscale Pixel Space

$$\mathbb{R}^{n=|\mathbf{Height}|} \times \mathbb{R}^{n=|\mathbf{Width}|}$$



Flatten row/column wise

$$\mathbb{R}^{n=|\mathbf{Height}| \times |\mathbf{Width}|}$$



"What if we have colored pixel space?"

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k -Nearest Neighbor: Image (MNIST Digits)

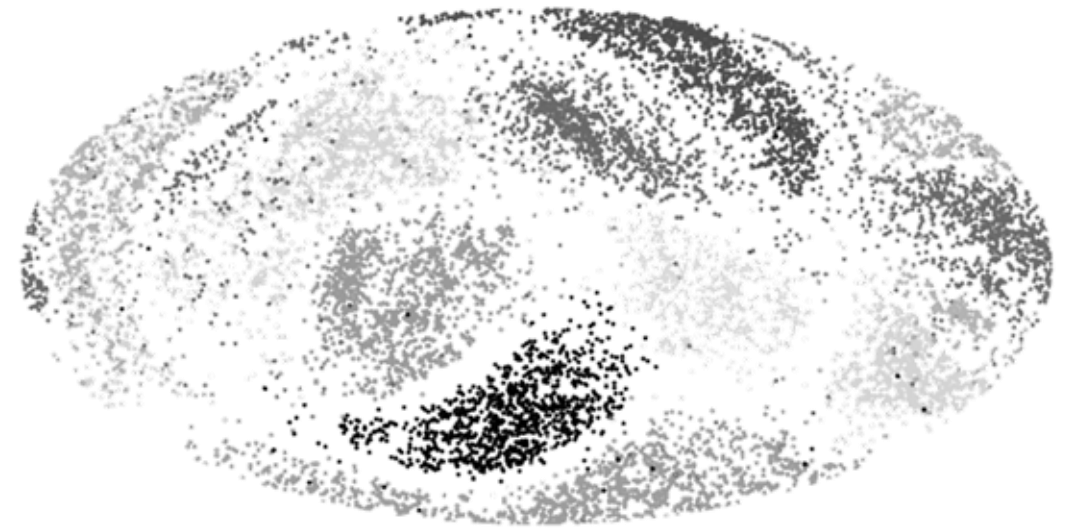
2022: Hyperspherical Variational Auto-Encoders, Davidson et al.

<https://arxiv.org/abs/1804.00891>

<https://github.com/nicola-decao/s-vae-pytorch>



(a) $\mathbb{R}^2$ latent space of the $\mathcal{N}$ -VAE.



(b) Hammer projection of S^2 latent space of the $\mathcal{S}$ -VAE.

0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9
 0 1 2 3 4 5 6 7 8 9

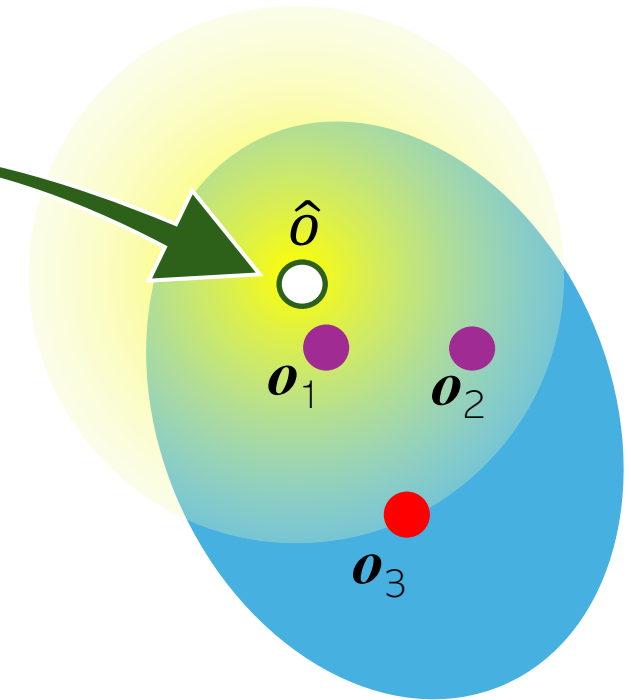
Figure 2: Latent space visualization of the 10 MNIST digits in 2 dimensions of both $\mathcal{N}$ -VAE (left) and $\mathcal{S}$ -VAE (right). (Best viewed in color)

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k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction

Person (Domain)	Animal (Domain)	Object (Domain)
Age ($\mathbb{N}$)	Height ($\mathbb{R}^+$)	Height ($\mathbb{R}^+$)
Height ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)
Gender ($\{M, F, N\}$)	Speed ($\mathbb{R}^+$)	Length ($\mathbb{R}^+$)
Occupation ($\{Student, Teacher, \dots\}$)	Type ($\{Mammal, \dots\}$)	Material ($\{Wood, Metal, \dots\}$)
Education ($\{BSc, MSc, \dots\}$)	Legs ($\mathbb{N}$)	Color ($\{White, Black, \dots\}$)
Experience ($\mathbb{N}$)	Lifespan ($\mathbb{N}$)	...
Location ($\{Canada, US, \dots\}$)	...	
Skill Level ($\{Novice, Experience, \}$)		
Income ($\mathbb{N}$)		
...		

 $e(o)$ 

Heterogenous Space
Discrete Space

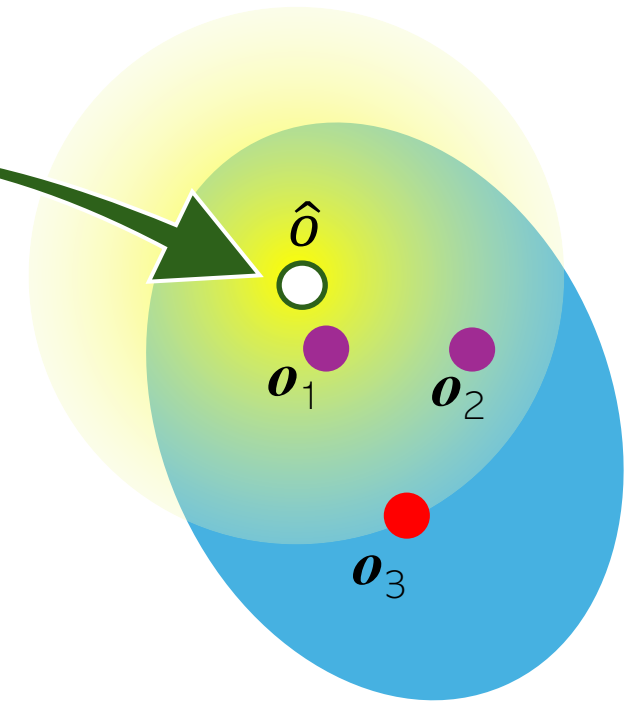
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k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction $\rightarrow$ Feature Scaling

Person (Domain)	Animal (Domain)	Object (Domain)
Age ($\mathbb{R}^+$)	Height ($\mathbb{R}^+$)	Height ($\mathbb{R}^+$)
Height ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)
Gender ($\{0.0, 1.0, 2.0\}$)	Speed ($\mathbb{R}^+$)	Length ($\mathbb{R}^+$)
Occupation ($\{0.0, 1.0, \dots\}$)	Type ($\{0.0, 1.0, \dots\}$)	Material ($\{0.0, 1.0, \dots\}$)
Education ($\{0.0, 1.0, 2.0\}$)	Legs ($\mathbb{R}^+$)	Color ($\{0.0, 1.0, \dots\}$)
Experience ($\mathbb{R}^+$)	Lifespan ($\mathbb{R}^+$)	...
Location ($\{0.0, 1.0, 2.0\}$)	...	
Skill Level ($\{0.0, 1.0, 2.0\}$)		
Income ($\mathbb{R}^+$)		
...		

$e(o)$



Homogeneous Continuous Space

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k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction $\rightarrow$ Feature Scaling

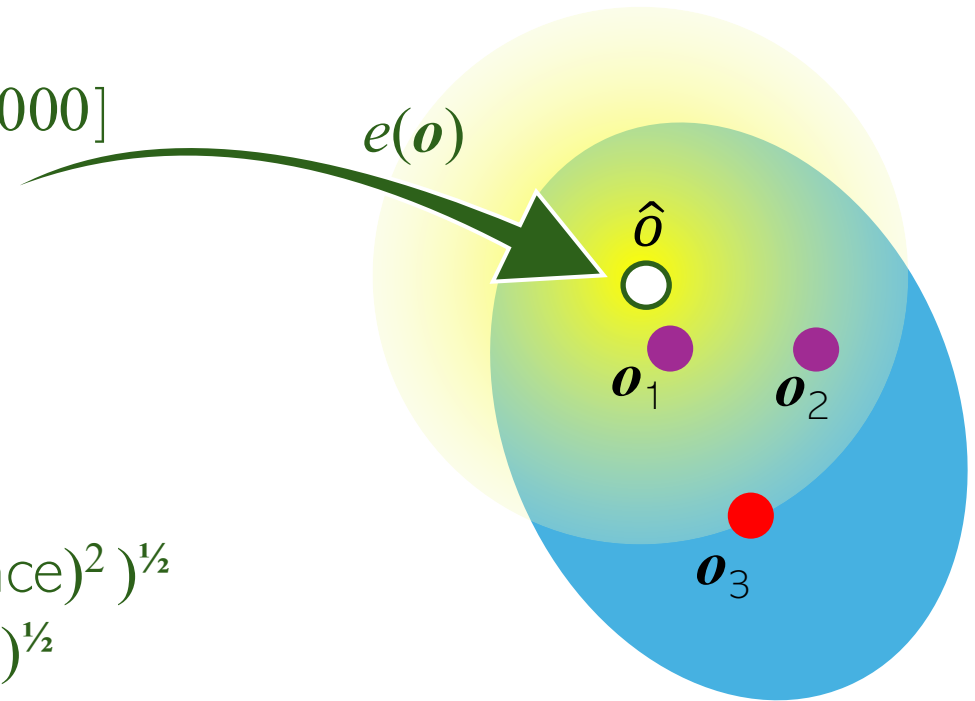
Age: $[1, 2, \dots, 100]$, Income: $[\$40,000, \dots, \$200,000]$

$e(\hat{o}: \text{Reia}) = (20, \$100,000)$

$e(o: \text{Hossein}) = (50, \$150,000)$

Euclidean Distance:

$$\begin{aligned} d(e(\hat{o}), e(o)) &= ((\text{age difference})^2 + (\text{income difference})^2)^{\frac{1}{2}} \\ &= ((20 - 50)^2 + (100,000 - 150,000)^2)^{\frac{1}{2}} \\ &= ((30)^2 + (50,000)^2)^{\frac{1}{2}} \end{aligned}$$



Income dimension dominates and overwhelms the age dimension.

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k-Nearest Neighbor: Entities

Hands-Crafted Feature Extraction → Feature Scaling

Min-Max Normalization: $x' = \frac{x - \min}{\max - \min}$

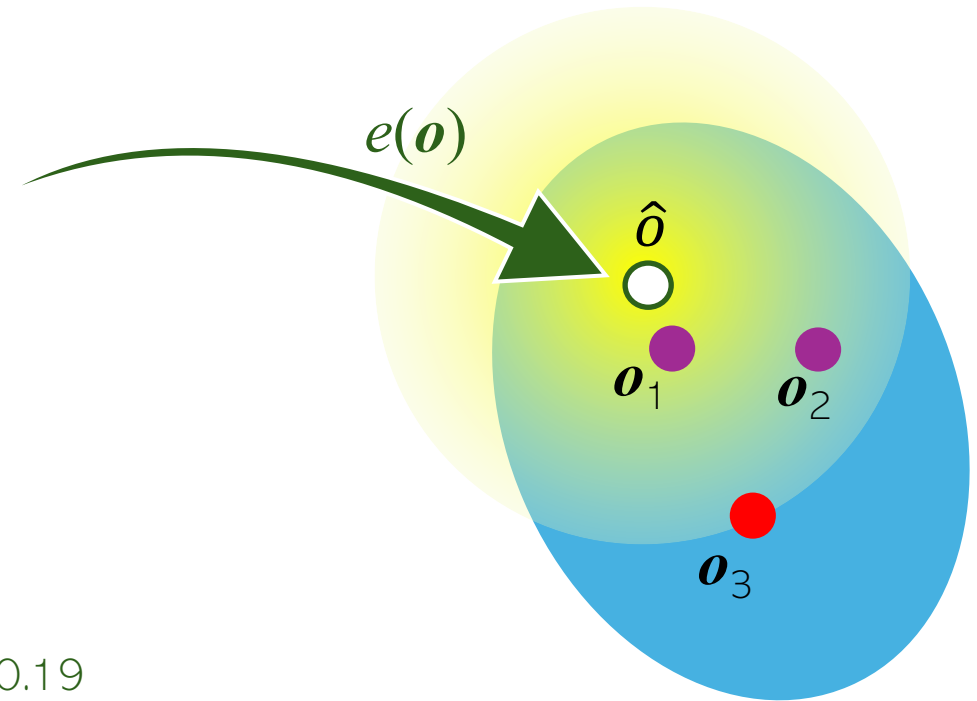
Age: [1, 2, ..., 100], Income: [\$40,000, ..., \$200,000]

$e(\hat{o}: \text{Reia}) = (20, \$100,000)$

$e(o: \text{Hossein}) = (50, \$150,000)$

$\text{Min}_{\text{Age}} = 1, \text{Max}_{\text{Age}} = 100$, then Reia's age become $x' = \frac{(20-1)}{100} = 0.19$

$\text{Min}_{\text{Income}} = \$40,000, \text{Max}_{\text{Income}} = \$200,000$, then Reia's income become $x' = \frac{(\$100,000 - \$40,000)}{\$200,000} = 0.30$



k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction $\rightarrow$ Feature Scaling

Standard Normal Distribution: $x' = \frac{(x - \mu)}{\sigma}$
(z-score)

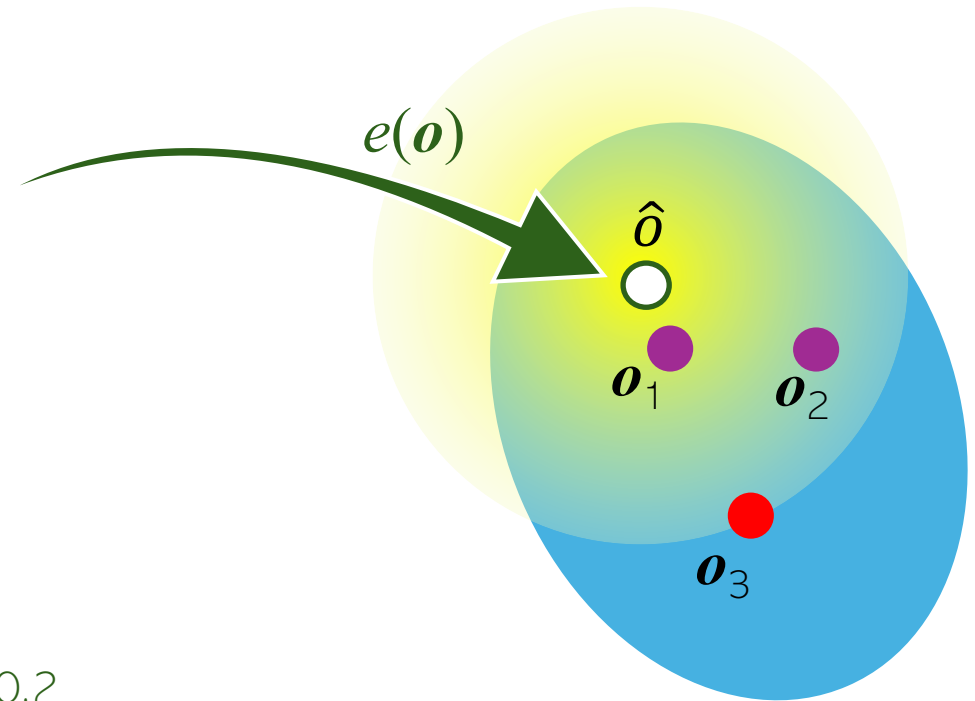
Age: [1, 2, ..., 100], Income: [\$40,000, ..., \$200,000]

$e(\hat{o}: \text{Reia}) = (20, \$100,000)$

$e(o: \text{Hossein}) = (50, \$150,000)$

$\mu_{\text{Age}} = 40$, $\sigma_{\text{Age}} = 10$, then Reia's age becomes $x' = \frac{(20 - 40)}{10} = -0.2$

$\mu_{\text{Income}} = \$80,000$, $\sigma_{\text{Income}} = \$20,000$, then Reia's income become $x' = \frac{(\$100,000 - \$80,000)}{\$20,000} = +1.0$



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 k -Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$

Can be a multi-label classifier? Justify your answer.

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 k -Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$

Is it explainable (xAI)?

Can it explain, justify, or provide evidence on why it picks a specific label?

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k -Nearest Neighbor: Pros & Cons

Pros:

Very simple and intuitive

Easy to implement

No explicit model assumptions like what would be voting strategy, distance function, ...

Naturally handles '*nonlinear*' decision boundaries (we cannot understand this now)

Useful baseline

Cons:

No training but at the cost of very expensive inference (linear to the training set size)

Requires storing training data

Sensitive to the choice of k , embedding space, distance function, feature scaling, ...

Poor in high dimensions (curse of dimensionality)

Outline: W05

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- IIII) Evaluation